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Hettangian and Sinemurian magnetostratigraphy from Paris Basin

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Abstract. A long core (1110 m) drilled at Montcornet (northeastern Paris Basin) provides early Jurassic magnetostratigraphic data coupled with biochronological control. About 600 paleomagnetic samples were obtained from a 148-m-thick series of Hettangian and Sinemurian rocks. A composite demagnetization using thermal (up to 300°C) followed by alternating field technique (up to 100 mT) is used to separate the magnetic components. A low unblocking temperature component (<250°C) with an inclination of about 65° is interpreted as a present-day field overprint. The characteristic remanent component with both normal and reversed antipodal directions is then isolated between 5 and 70 mT. Eighty-one polarity intervals are recognized in this study. The higher reversal frequency of the late Hettangian/early Sinemurian time interval contrasts with a lower reversal rate observed in the rest of the early Liassic. A rough mean estimate of about 5 reversals/m.y. can be proposed for the earliest Jurassic. These results represent a significant contribution to the magnetic polarity reversal timescale for a time interval hitherto poorly known and add to the magnetic reversal frequency curve of the last 350 m.y.

Introduction

For postlate Jurassic times, correlations between magnetostratigraphic polarity intervals and marine magnetic anomalies have been established [e.g., *Lowrie*, 1989]. These have been largely used for the construction of global scale stratigraphic correlations and have immediate applications for the study of the geomagnetic field behavior and geodynamic processes. As the record of oceanic magnetic lineations does not extend beyond late Jurassic time, magnetostratigraphy is the only method for investigating older changes in geomagnetic polarity. During the last decade, several magnetostratigraphic data were obtained for the Jurassic [*Horner and Heller*, 1983; *Channell et al.*, 1984; *Steiner et al.*, 1987, *Steiner et Ogg*, 1988; *Galbrun et al.*, 1988, 1990; *Witte and Kent*, 1990; *Juárez et al.*, 1994], the Triassic [*Heller et al.*, 1988; *Steiner et al.*, 1989; *Witte and Kent*, 1990; *Ogg and Steiner*, 1991; *Gallet et al.*, 1992, 1993, 1994], and the late Permian [*Haag and Heller*, 1991; *Heller et al.*, 1995). Despite superimposition of high frequency variations, these results document an increasing trend of the reversal rate from the end-Kiaman reversed superchron to late Jurassic time. This pattern of change in reversal frequency compares to the one that followed the long normal chron of late Cretaceous age (*Gallet*



Figure 1. Previous Hettangian and Sinemurian magnetostratigraphic sequences from Hungary (BC, Bakony, Transdanubian Central Mountains [*Marton et al.*, 1980]), Italy and Switzerland (AT, Alpe Turati; BR, Breggia [*Horner and Heller*, 1983]), Italy, Umbrian Apennines (FA, Fonte Avellana; CI, Cingoli [*Channell et al.*, 1984]), and the United States (NB, Newark Basin [*Kent et al.*, 1995]). Absolute timescale after *Harland et al.* [1990].

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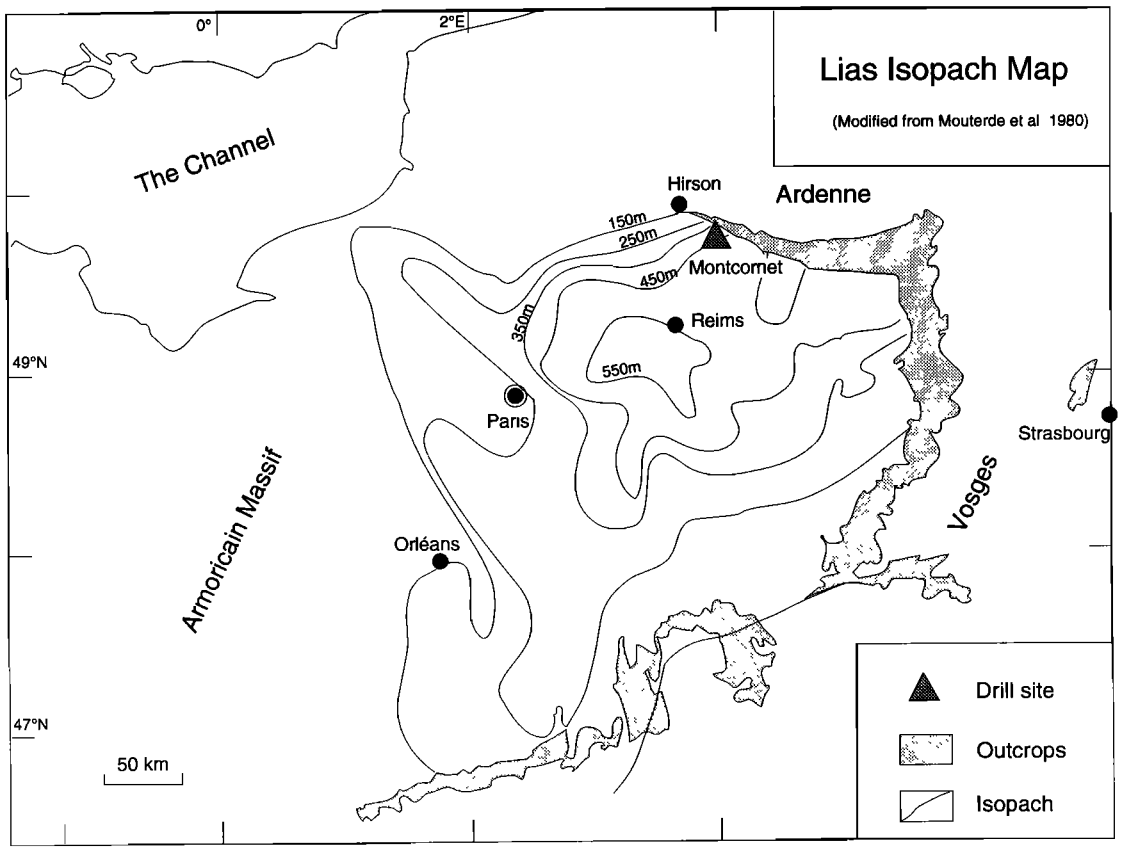


Figure 2. Lias isopach map of the Paris Basin [after *Mouterde et al.*, 1980].

et al., 1992). Therefore *McFadden and Merrill* [1984] and *Courtillot and Besse* [1987] postulated a 150-200 m.y. time constant in the reversal process for the last 350 m.y.

However, the detailed pattern of reversal frequency of some Mesozoic time intervals such as the early Jurassic remains poorly known. First attempts in establishing early Jurassic magnetostratigraphy were made by *Channell et al.* [1984] in the Umbrian Apennines, Italy, by *Marton et al.* [1980] in the Transdanubian Central Mountains, Hungary, and by *Witte et al.* [1991] in the Newark Basin, United States. The results of these contributions are summarized in Figure 1. They suggest a low average reversal rate: six polarity intervals for the 9 m.y. duration of the Sinemurian [*Channell et al.*, 1984] and an essentially normal polarity interval including a single short reversed interval for the earliest Hettangian [*Kent et al.*, 1995]. *Channell et al.* [1984] also mentioned problems pertaining to their Sinemurian section such as low sedimentation rate, gaps, and insufficient biochronological control.

A 1120-m-long core drilled near Moncornet in northeastern Paris Basin (49.7°N, 4°E) by Agence Nationale pour la gestion des Déchets Radioactifs (ANDRA) in 1989 provided an opportunity to study the magnetic polarity sequence of early Jurassic times. The uninterrupted polarity succession presented in this paper covers the Hettangian and Sinemurian stages, which are represented by a 148 m-thick series of epicontinental marine rocks. This long core permits an almost continuous paleomagnetic sampling of unweathered marine rocks. In addition to the foraminifera and ostracod age data provided by P. Andreieff (Bureau de Recherches Géologiques et Minières, unpublished report, 1991), this study provides further age constraints based on ammonoids and pollens.

Stratigraphic Framework

The drilling locality is paleogeographically situated on the northern side of the early Liassic transgressive embayment of Paris Basin (Figure 2). Core orientation is nearly vertical (280°/84°) as measured at 920 m depth using an inclinometer. Regional bedding attitude is almost horizontal (180°/0.5°) thus implying minimal bedding correction.

Summary of the Lithological Succession

At 1075 m depth, transgressive open marine rocks of early Jurassic age comformably overlie late Triassic (Rhaetian?) nonmarine variegated silty clays and marls. These oldest marine rocks consist of about 70 m of light to medium grey, impure, cross-bedded bioclastic limestone with marly intercalations. They are broadly interpreted as intertidal depositional environments. Gradational dark grey, biotubated silty marl and clays with subordinate impure bioclastic limestone are found around 1005-m depth. Macrofauna are mainly represented by unabraded benthonic forms such as the bivalve *Gryphea. arcuata*. This change in lithology reflects a transition to a deeper, essentially subtidal depositional environment. Around 940 m depth, the rocks grade upward into dark grey to black, occasionally bioturbated silty carbonaceous shales suggestive of an oxygen-deficient, offshore depositional setting. In this upper part of the

Table 1a. Palynological Data

Sample Depth	1074.9		1074.75		1074.23	1073.18		1070.01		1069.60	1068.10		1067.81	
	N	%	N	%	N %	N	%	N	%	N %	N	%	N	%
Total number of counted microfossils	117		98		26 SQ	119		113		27 SQ	148		124	
Total number of spores	55	47	57	58	13	31	26	44	39	11	34	23	60	48
Total number of pollens	28	24	37	38	8	48	40	51	45	13	83	56	45	36
Total number of acritarchs	32	27	1	1	3	40	34	17	15	3	29	20	16	13
Total number of dinoflagellates	1	0.9	2	2	1	0		0		0	0		1	0.8
Total number of algae	0		0		0	0		1	0.9	0	1	0.7	1	0.8
Total number of foraminiferas	1	0.9	1	1	1	0		0		0	1	0.7	1	0.8
Total number of marine microfossils	34	29	4	4.1	5	40	34	18	16	3	31	21	19	15
Total number of continental microfossils	83	71	94	96	21	79	66	96	84	24	117	79	105	85
Spores														
Apiculatisporites leviomatus	0		0.5		1	0		1	0.9	1	0		0	
Auritulinasporites spp.	0		0		0	0		0		0	0.5		1	0.8
Calamospora spp.	1	0.9	1	1	0	0		0.5		0	0		0	
Chasmatosporites sp. A	0		0		0	0		0		0	0		0.5	
Conbaculatisporites mesozoicus	0.5		0.5		0	0		0		0	0		0	
Concavisporites torus	1	0.9	1	1	1	2	1.7	3	2.7	1	0		0.5	
Concavisporites juriensis	0		0		1	0		0		0	0		0	
Convolutispora microrugulata	0		0.5		1	1	0.8	0.5		0	0		0	
Deltoidospora spp.	10	8 5	4	4	1	5	4.2	5	4.4	1	4	2.7	21	17
Echinatisporites sp.	1	0.9	0		0	0		0		0	0		0	
Heliosporites reissingeri	8	6.8	23	23	1	4	3.4	3	2.7	2	7	4.7	9	7 3
Iraquispora lavigata	0		0		1	0		0		0	0		0	
Leptolepidites argenteaeformis	32	27	24	24	3	13	11	28	25	3	15	10	18	15
Leptolepidites sp. A	0		0.5		0	0		0		0	0		0	
Lycopodiacidites rugulatus	0.5		1	1	1	2	1.7	0.5		1	1	0.7	3	2 4
Lycopodiumsporites austroclavatidites	0.5		0.5		0	0.5		0		1	0		0.5	
Lygodiumsporites adriennis	1	0.9	2	2	1	2	1.7	4	3.5	1	5	3.4	8	6.5
Lygodiumsporites adriennis minor	0.5		0		0	0		0		0	0		0	
Nevesisporites bigranulatus	0.5		0.5		0	0.5		0.5		0	0		0.5	
Nevesisporites limatulus	0		0		0	0		0.5		0	0		0	
Osmundaciidites weilmanii	0		0		0	0		0		0	0.5		0	
Retitriletes semireticulatus	0		0		0	0		0		0	0.5		0	
Retusisporis mesozoicus	0		0		0	1	0.8	0		0	0		0	
Ricciisporites tuberculatus	0		0.5		0	0		0		0	0		0	
Schizosporis spp.	0		1	1	0	0		0		0	0.5		0	
Stereisporites spp.	0		0.5		1	0		0.5		0	1	0.7	0.5	
Todisporites major	1	0.9	0		0	0		0		0	0		0	
Trilites microverrucosus	0		0		0	1	0.8	0		0	0		0	
Trachysporites fuscus	0		1	1	0	0.5		0.5		0	0		0	
Pollens	0													
Araucariacites australis type	0		0.5		0	0		0		0	0		0	
Classopollis spp. (groupe)	7	6	15	15	3	33	28	33	29	5	48	32	25	20
Disaccates indifférentiés	10	8.5	7	7	1	3	2.5	5	4.4	3	15	10	13	10
Inaperturopollenites hiatus	11	9.4	15	15	1	12	10	10	8.8	2	15	10	7	5 6
Monosulcites spp.	0		0		1	0		3	2.7	1	3	2	0.5	
Perinopollenites elatoides	0		0		1	0		0.5		1	1	0.7	0	
Quadraeculina anellaeformis	0		0		0	0		0		0	0		0.5	
Vitreisporites pallidus	0		0.5		1	0.5		0		1	1	0.7	0	
Marine Microfossils	0													
Baltisphaeridium spp.	0		0		0	5	4.2	0		0	0		0	
Leiofusa jurassica	0		0		0	0		0		0	2	1.4	1	0.8
Micrhystridium lymensis	32	27	1	1	3	30	25	17	15	3	25	17	15	12
Micrhystridium spp.	0		0		0	5	4.2	0		0	0		0	
Micrhystridium stellatum	0		0		0	0		0		0	1	0.7	0	
Veryhachium triepinosum	0		0		0	0		0		0	1	0.7	0	
Tytthodiscus spp.	0.5		0		0	0		1	0.9	0	0		1	0.8
Valvaeodinium koessenium	0		0		0	0		0		0	0		0	
Beaumontella delicata	0		0		0	0		0		0	0		0	
Dapcodinium priscum	0.5		2	2	1	0		0		0	1	0.7	1	0.8
Dini A	0.5		0		0	0		0		0	0		0	
Dini B	0.5		0		0	0		0		0	0		0	
Indeterminate dinoflagellates	1	0.9	0		0	0		0		0	0		0	
Organic tests of foraminiferas	2	1 7	1	1	1	0		0		0	1	0 7	1	0 8
Sample depth	1074 90		1074.75		1074.23	1073.18		1070.01		1069.60	1068 1		1067 81	

Number of counted microfossils is N. and percentage. Rare species not counted (indicated as 0.5 in column N) hence corresponding percentage (%) is not calculated. When specimens are poorly preserved, semi-quantitative estimates are given in column N as follows. rare(1), common (2), abundant (3), dominant (4).

Table 1a. (continued)

1056.27		1055 76		1054.49	1054 06		1047 28		1046.66		1045.68		1039.63		1039.08	
N	%	N	%	N	N	%	N	%	N	%	N	%	N	%	N	%
107		15		12 SQ	147		145		139		117		138		143	
11	10	6	40	5	20	14	5	3.4	4	2.9	5	4 3	4	2.9	4	2.8
43	40	7	47	5	104	71	90	62	80	58	86	74	98	71	92	64
50	47	2	13	1	23	16	50	34	55	40	25	21	35	25	46	32
2	1 9	0		1	0		0		0		1	0.9	0		1	0.7
1	0.9	0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		1	0.7	0	
53	50	2	13	2	23	16	50	34	55	40	26	22	36	26	47	33
54	50	13	87	10	124	84	96	66	84	60	91	78	102	74	96	67
0.5		0		0	0.5		1	0.7	0		1	0.9	0.5		0.5	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0.5		0.5		1	0.7	0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0 5		1	6.7	1	0		4	2.8	3	2.2	3	2.6	0.5		3	2.1
0		0		0	0		0		0		0		0		0	
4	3.7	1	6.7	1	4	2.7	0		0		1	0.9	3	2.2	1	0.7
0		0		0	0		0		0		0		0		0	
3	2.8	1	6.7	1	12	8 2	0.5		1	0.7	0.5		0		0	
0		0		0	0		0		0		0		0		0	
0		0		1	3	2	0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
4	3.7	1	6.7	1	0		0		0		0		0.5		0.5	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0.5		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0.5		0		0	
0		1	6.7	0	1	0.7	0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		1	6.7	0	0		0		0		0.5		0.5		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0.5		0		0.5		0		0.5		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
30	28	5	33	1	60	41	75	52	43	31	40	34	75	54	40	28
8	7 5	1	6.7	1	37	25	10	6 9	15	11	20	17	8	5.8	15	10
5	4 7	1	6.7	1	7	4.8	5	3 4	15	11	25	21	15	11	33	23
0		0		1	0		0 5		1	0 7	0		0.5		0	
0		0		1	0 5		0		5	3.6	0		0.5		0	
1	0 9	0		0	0 5		0		0		0		0		0	
0		0		0	0		0		1	0.7	1	0.9	0.5		0.5	
0		0		0	0		0		0		0		0		0.5	
0		0		0	0		0		0		0		0		0	
50	47	2	13	1	23	16	50	34	55	40	25	21	35	25	45	31
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0.5	
1	0 9	0		0	0		0		0		0		0		0	
2	1.9	0		1	0		0		0		0		0		1	0 7
0		0		0	0		0		0		0		0		0.5	
0		0		0	0		0		0		1	0.9	0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0		0	
0		0		0	0		0		0		0		0 5		0	
1056 27		1055 76		1054.49	1054 06		1047 28		1046 65		1045 58		1039.63		1039.08	

Table 1b. Palynological and Sequence Stratigraphy Ages

Parameter	Ages	Depth, m
Palynological age	Early/middle Hettangian	1074.9 - 1067.8
	middle Hettangian to early Sinemurian	1056.3 - 1039.1
Sequence stratigraphy	MFSRh3? Planorbis	1074.2
	SBHe1? Portlocki	1073.2 - 1056.3
	MFSHe1? Laqueus	1054.5
	SBHe2? Complanata	1053.5
	MFS He2? Rotiforme	1039.6 - 1039.1
Palynological age and sequence stratigraphy	early Hettangian	1074.9 - 1073.2
	early Hettangian to middle Hettangian	1070.0 - 1067.8
	middle Hettangian	1056.3 - 1053.5
	late Hettangian to early Sinemurian	1053.5 - 1039.1

Table 2. Ammonoid and Bivalve Occurrences Between 920 and 992 m Depth

Depth, m	Taxa	Age
		Early Carixian
Jamesoni Zone		
920.60	*Polymorphites* sp.	
		Late Sinemurian
Raricostatum Zone		
Macdonnelli Subzone:		
927.82	*Leptechioceras* sp	
929.65	*Echioceratidae* sp.	
929.90	*Leptechioceras* (*Neomicroceras*) aff. *commune*	
931.30	*Echioceras* sp. nov.	
931.43	*Echioceratidae* sp.	
931 45	*Leptechioceras* (*Neomicroceras*?) sp	
Raricostatum Subzone		
931.74	*Echioceras* juv.	
932.07	*Echioceratidae* juv.	
932.13	*Echioceras* gr. *raricostatum*	
932.20	*Gryphaea* aff. *cymbium**	
	Echioceras sp. juv.	
	Paltechioceras ? cf. *boehmi*	
932.22	*Echioceras* gr. *raricostatum*	
932.23	*Echioceras* gr. *raricostatum*	
932.35	*Echioceras* juv. (*E.* gr. *raricostatum* ?)	
932.26	*Echioceras* gr. *raricostatum*	
932.44	*Echioceras raricostatum*	
932.48	*Echioceras raricostatum*	
932.75	*Echioceras raricostatum*	
933.32	*Echioceras* sp. juv.	
Densinodulum Subzone		
934.85	*Hemmimicroceras* sp. juv.	
935 67	*Hemmimicroceras* aff. *subplanicosta*	
935.55	*Hemmimicroceras* cf. *subplanicosta* juv.	
935.86	*Oxynoticeratidae* sp. juv.	
	Hemmimicroceras aff. *subplanicosta*	
936.33	*Hemmimicroceras subplanicosta*	
937.07	*Hemmimicroceras* aff. *subplanicosta*	
937.15	*Hemmimicroceras* cf. *subplanicosta*	
937.26	*Hemmimicroceras* sp. juv.	
937.57	*Hemmimicroceras* sp. juv.	
Obtusum Zone		
939.80	*Asteroceras* sp.	
943.36	*Gryphaea cymbium* *	
	Xipheroceras gr. *planicosta* juv.	
943.96	*Xipheroceras* juv.	
945.10	*Aegasteroceras* sp.	
948.45	*Asteroceras* sp.	
948.88	*Promicroceras* gr. *planicosta*	
		Early Sinemurian
992 50	*Gryphaea* gr. *arquata* *	

* Bivalve

section, macrofossils (cephalopods and thin-shelled bivalves) are essentially nektonic and pseudoplanktonic.

Age Control

Palynological dating and interpretation of both lithostratigraphic succession and gamma ray profile in terms of sequence stratigraphy (R. Jan du Chêne, unpublished report, 1994, see Table 1) lead to the following age assignments: (1) 1074.9 to 1073.2 m, early Hettangian; (2) 1070.0 to 1067.8 m, early or middle Hettangian; (3) 1056.3 to 1053.5 m, middle Hettangian; and (4) 1053.5 to 1039.1 m, late Hettangian to early Sinemurian, undifferentiated.

Although no obvious unconformity was observed at the Triassic-Jurassic boundary, it could exist because uppermost Triassic strata did not yield enough well-preserved floras. A Rhaetian age for strata below 1075 m is based on regional facies correlation.

Ammonoids and bivalves lead to the following age assignments (see details in Table 2): (1) 992.5 m, early Sinemurian; (2) 948.9 to 939.8 m, early late Sinemurian, Obtusum Zone; (3) 937.6 to 927.8 m, late Sinemurian, Raricostatum Zone; and (4) 920.6 m, early Carixian, Jamesoni Zone.

The lowermost ammonoid occurrence (Obtusum Zone, early late Sinemurian) approximately coincides with the transition to oxygen-deficient, offshore depositional environment. Comparison with the revised standard scale [*Groupe Français d'étude du Jurassique*, 1996] shows that diagnostic ammonoids indicative of the uppermost Obtusum and Oxynotum zones are missing between 939.80 and 937.57 m.

Ostracods and foraminiferas (P. Andreieff, Bureau de Recherches Géologiques et Minières, unpublished report, 1991), suggest that the Hettangian-Sinemurian boundary should be placed between 1023 and 1035 m. (last appearance datum of *Klingerella translusens* [*Donze*, 1985], see Figure 3).

The biostratigraphy suggests that deposition initiated by the early Hettangian. The position of the Hettangian-Sinemurian boundary could not be precisely established because of the discrepancy of the paleontologic age estimates (Figure 4). On the one hand, ostracods and foraminiferas suggest that the boundary occurs between 1023 and 1035 m. On the other hand, palynological data suggest that the boundary should be placed between 1039.8 and 1053.5 m. As this indetermination cannot yet be resolved, a conservative estimate leads to placement of the boundary within the maximum error interval,

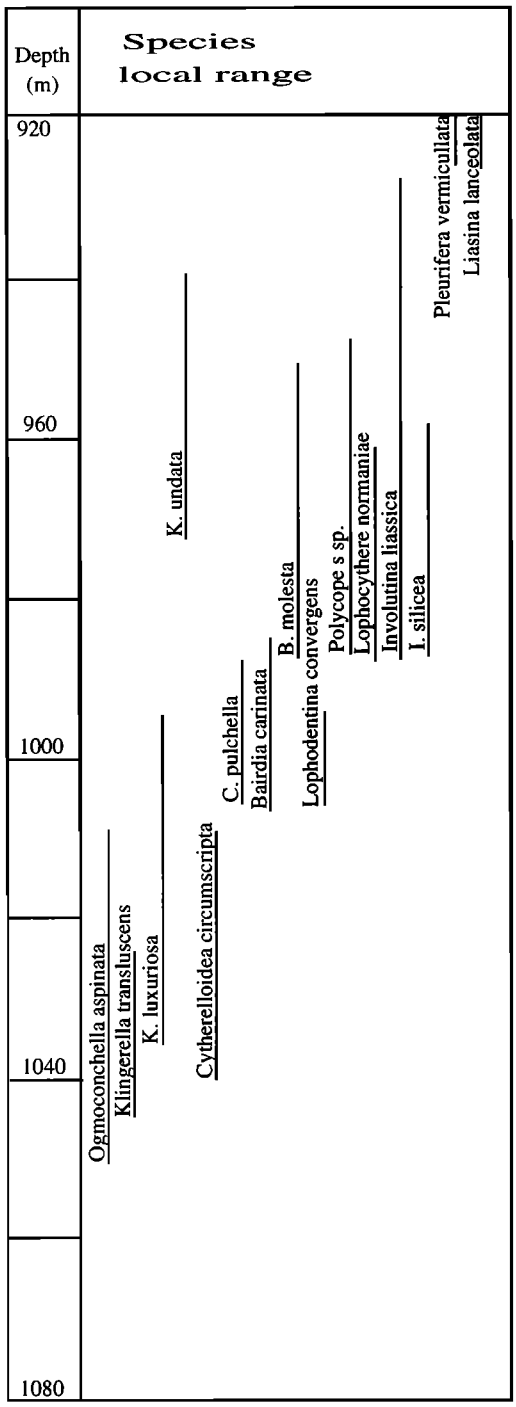


Figure 3. Micropaleontology local range.

that is between 1023 and 1053.5 m. Macrofossil and microfossil occurrences bracket the early-late Sinemurian boundary between 992.5 and 987 m and the Sinemurian-Carixian stage boundary around 927 m.

Magnetostratigraphic Sampling

Core segments of 11 cm in diameter were recovered with a 3-m-long drill string. Length of recovered core segments without internal rotation varies from 10 cm to 9 m. Usually, contacts between unbroken successive 3-m-length segments were frequently preserved in mechanically resistant rocks thus making it possible to restore composite core segments (CCS) as long as 10 m without internal rotation. An arbitrary generating line was marked on each segment, either single or composite. Standard paleomagnetic samples were cored perpendicularly along the generating line of each segment. The sample coordinates are defined with respect to this arbitrary generating line for each CCS before vertical correction of core.

Clay-rich segments were sampled using compressed air, whereas carbonate segments were sampled using water as a coolant. The mean sampling interval was approximately 30 cm. In a subsequent phase of sampling, core interval was reduced to about 5 cm for polarity changes originally defined by single samples. About 600 oriented samples were cored between 1075 and 927 m thus covering the Hettangian and Sinemurian stages.

Rock Magnetic Study

Isothermal Remanent Magnetization (IRM) Acquisition and Thermal Demagnetization of the Three-Axes Composite IRM

Acquisition of IRM is an efficient method to determine the coercivity spectrum of magnetic minerals in a rock specimen. IRM acquisition curves from representative limestone and clay

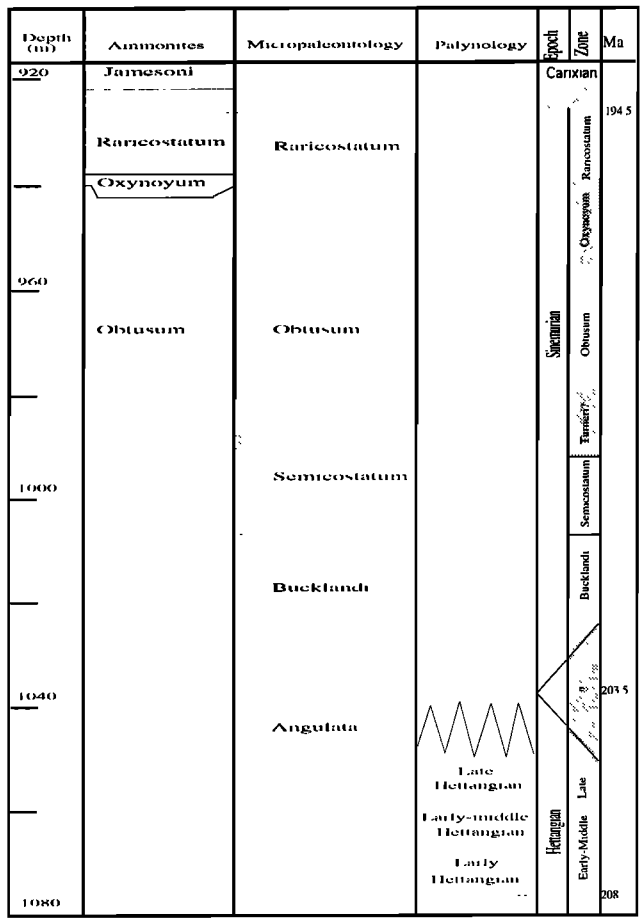


Figure 4. Age control and position of stage boundary from biostratigraphic data. Absolute timescale after *Harland et al.* [1990].

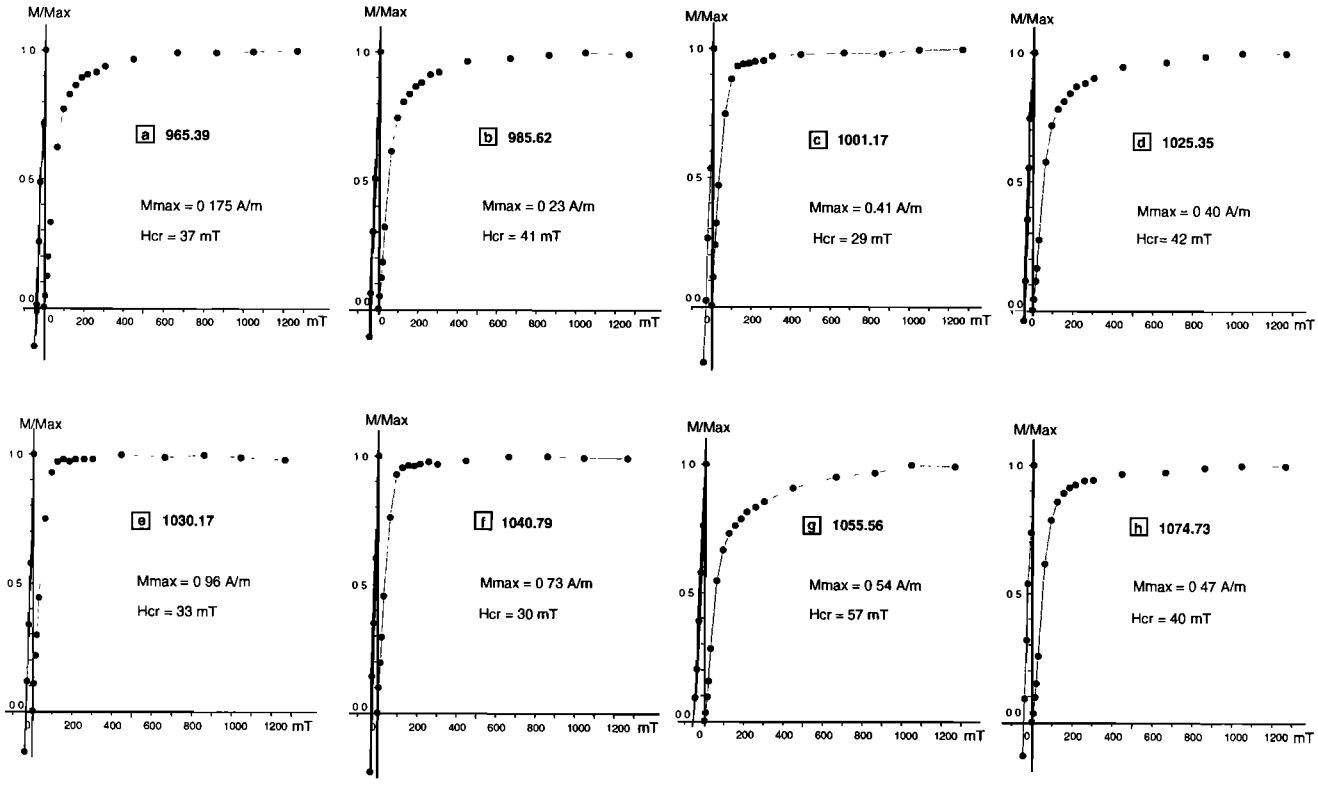


Figure 5. Normalized isothermal remanent magnetization (IRM) acquisition curves for eight representative samples. Note the rapid increase in magnetization up to 0.3 T and a maximum remanent coercivity around 40 mT for all specimens

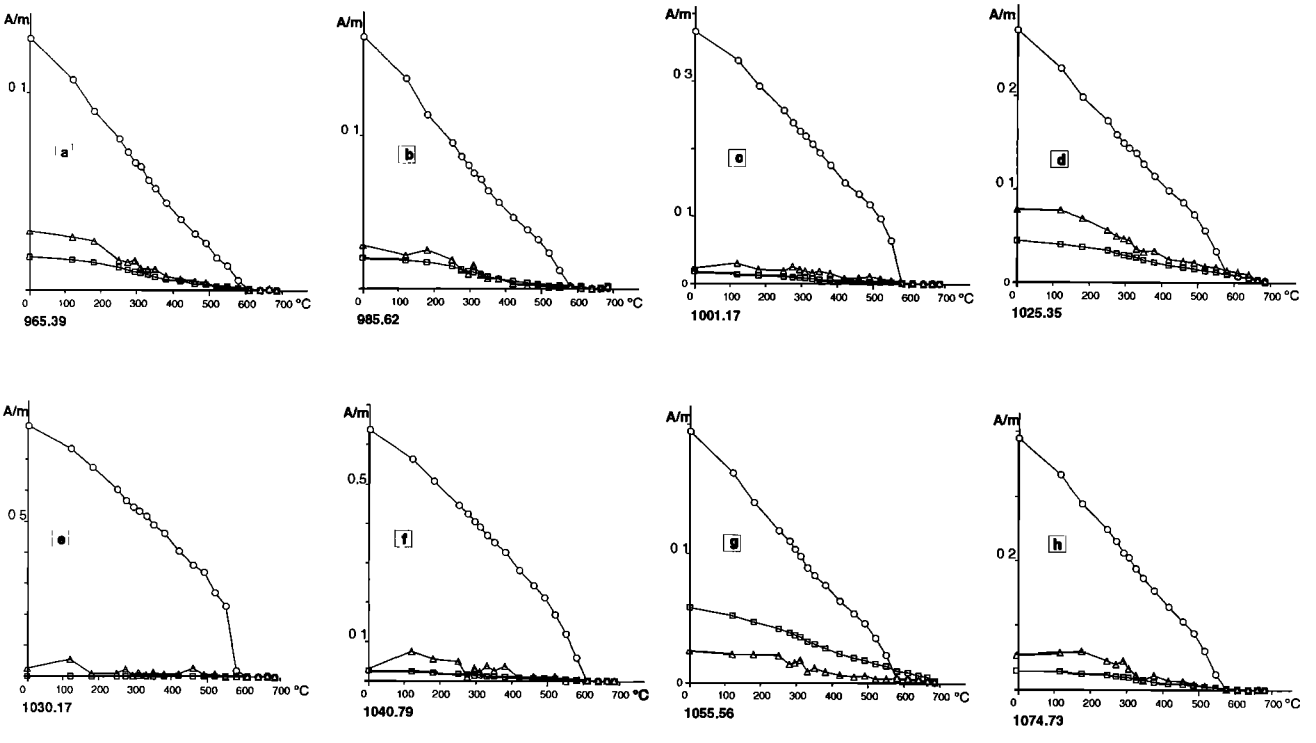


Figure 6. Three-component IRM thermal demagnetization [*Lowrie*, 1990]. Circles indicate soft component, 0.12T field; triangles indicate medium component, 0.4T field; and squares indicate hard component, 1.2T field. Soft component shows unblocking temperatures around 580°C in most specimens. Coexisting with magnetite is a high coercive component with unblocking temperatures of 680°C, most likely hematite.

samples indicate an initial rapid increase of the magnetization in a steady field up to 0.3 T, followed by a slowly increasing or constant magnetization up to 1.2 T (Figure 5). Bulk coercive remanent forces (Hcr) range between 30 and 47 mT which indicates the predominance of a low coercivity magnetic mineral such as magnetite. In most samples, the regular increase of magnetization beyond 0.3 T indicates the presence of at least some high-coercivity magnetic mineral, most probably fine-grained hematite.

Following the method described by *Lowrie* [1990], a composite IRM was then thermally demagnetized for a more accurate identification of magnetic carriers. Different magnetic fields (1.2, 0.4 and 0.12 T) were successively applied to each of the three perpendicular axies of a sample prior to thermal demagnetization. The low coercive component is dominant and usually corresponds to a maximum unblocking temperature of about 580°C (Figure 6). These results confirm that magnetite is the principal magnetic carrier, although a higher coercive mineral characterized by unblocking temperatures around 680°C (e.g., hematite) occurs in most samples (Figure 6). Since the high coercive component (1.2 T) comprises only 10% of the total composite IRM, the contribution of hematite to the remanent magnetization is very small.

Thin-Section Observation

Study of thin sections indicates that magnetite occurs as regular rounded grains (up to 40 μm in diameter) and is most likely of detrital origin (Y. Zinbi, personal communication, 1995). Macroscopic and microscopic examinations indicate pyrite and goethite are present in most samples. The pyrite has either a spheroidal or a cubic texture and probably replaces magnetite under reducing conditions [*Leslie et al.*, 1990]. Original magnetite is partially replaced by pyrite, which may in turn be replaced by goethite. This mineralogical transformation probably occurred soon after deposition. The various IRM tests failed to reveal the presence of goethite, because the necessary field intensity required to change magnetization of goethite may be as high as several Tesla [*Rochette and Fillion*, 1989]. Fluid circulation may generally account for the transformation of pyrite into irregular masses of goethite similar to that recognized in some samples. However, these rocks with a high-clay mineral content did probably not undergo significant postdiagenetic fluid circulation. We therefore assume that goethite formed during diagenesis soon after deposition. From a practical point of view, the presence of goethite was ignored in the magnetic analyses of the natural remanent magnetization (NRM) because of random distribution of the component measured between room temperature and 120°C.

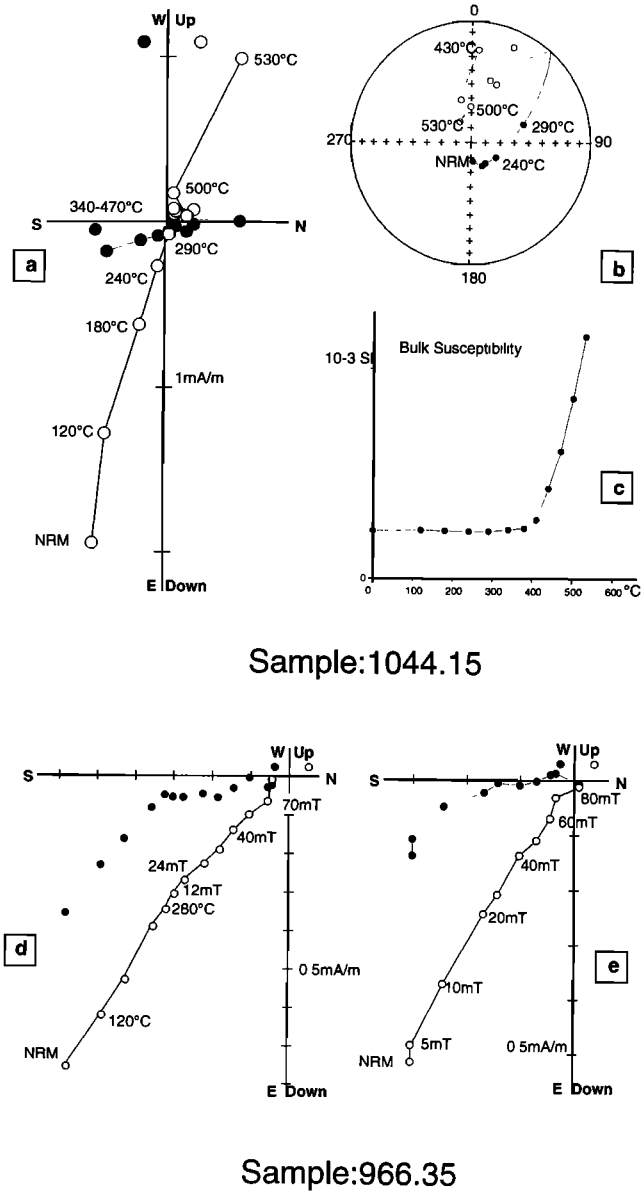


Figure 7. Examples of demagnetization. Thermal demagnetization is shown for specimen 1044.15. (a) Orthogonal vector projection is shown. (b) Equal-area projection is shown. Solid/open symbols refer to lower/upper hemispheres, respectively; horizontal coordinates are arbitrary; and vertical coordinate indicates drill core axis. (c) Bulk susceptibility change during heating is shown. Comparative demagnetization for specimen 966.35 is shown. (d) Orthogonal vector projection of the combined thermal and alternating field (AF) (mT is millitesla) technique is shown. (e) AF technique is shown.

Natural Remanent Magnetization (NRM) Analysis

Magnetic Measurements

The remanent magnetization of the samples was measured using a CTF cryogenic magnetometer in the shielded paleomagnetic laboratory at the Institut de Physique du globe de Paris. Sixty pilot samples were chosen for stepwise thermal demagnetization. The bulk magnetic susceptibility of the specimens was measured at each demagnetization step to check the stability of the magnetic minerals during heating.

NRM intensities range between 0.1 and 5 mA/m. A random component, completely removed at 120°C, is found in a vast majority of samples. A well defined (linear) lower unblocking temperature component is removed around 300°C (Figure 7a). Magnetization becomes generally unstable at higher temperatures, up to the Curie point of magnetite (Figure 7b). A rapid increase of the bulk magnetic susceptibility above 400°C (Figure 7c) indicates alteration of the magnetic mineralogy. Such change may result from oxidation of

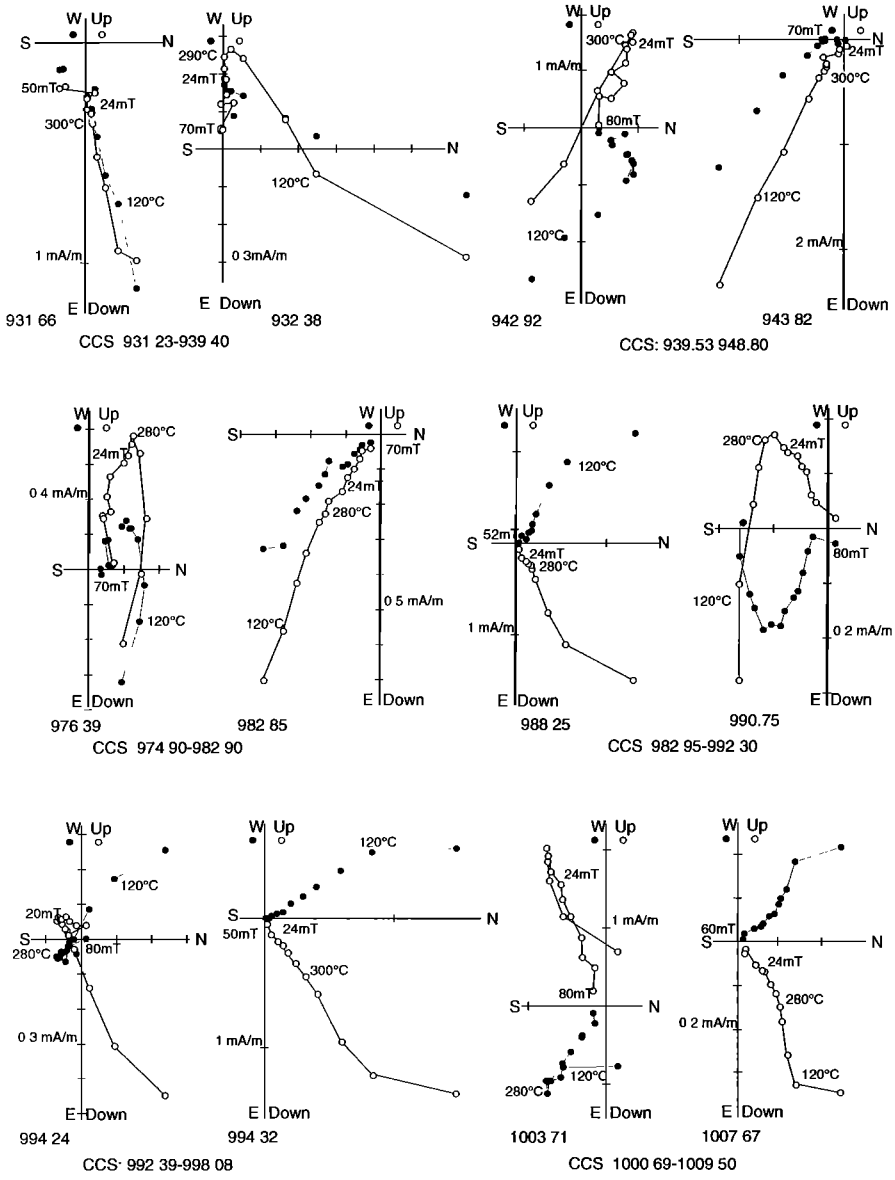


Figure 8. Orthogonal vector projection of the combined thermal and AF demagnetizations for representative specimens from twelve composite core segments (CCS), same convention as in Figure 7.

sulphides thus making isolation of the higher temperature component impossible. However, this problem is easily circumvented by switching to alternating field (AF) demagnetization technique (with a 3-axis AF demagnetizer coupled to a 2 G cryogenic magnetometer). AF demagnetization between 5 and 50 mT clearly indicates the presence of a low-coercivity component, which is likely to correspond to the former higher-temperature component. Only one component is removed during this second AF demagnetization phase, which suggests complete cleaning at 300°C and occurrence of two distinct components. In contrast, an exclusively AF demagnetization procedure shows simultaneous removal of the two components between 5 and 40 mT (figure 7e) thus impeding separation of the components (Figure 7d). Although the combined technique may not always resolve the two componants, the magnetic polarity is unambiguously identified. We therefore routinely used the hybrid demagnetization procedure by heating in five steps to 300°C, followed by progressive AF demagnetization to 60-90 mT.

Magnetic Interpretation

Demagnetization data were plotted on orthogonal vector diagrams (Figure 8) and directions were determined for each sample by principal component analysis [*Kirschvink*, 1980]. Mean directions were calculated using the statistical methods of *Fisher* [1953].

A first randomly oriented component is removed around 120°C. This component is probably acquired during drilling, storage, and specimen sampling.

For each composite core segment (CCS), α_{95} values of lower temperature component are generally smaller than 7°. Application of the *McFadden and Reid* [1982] method yields a mean inclination I = 62° ± 2°, a value which is in relatively

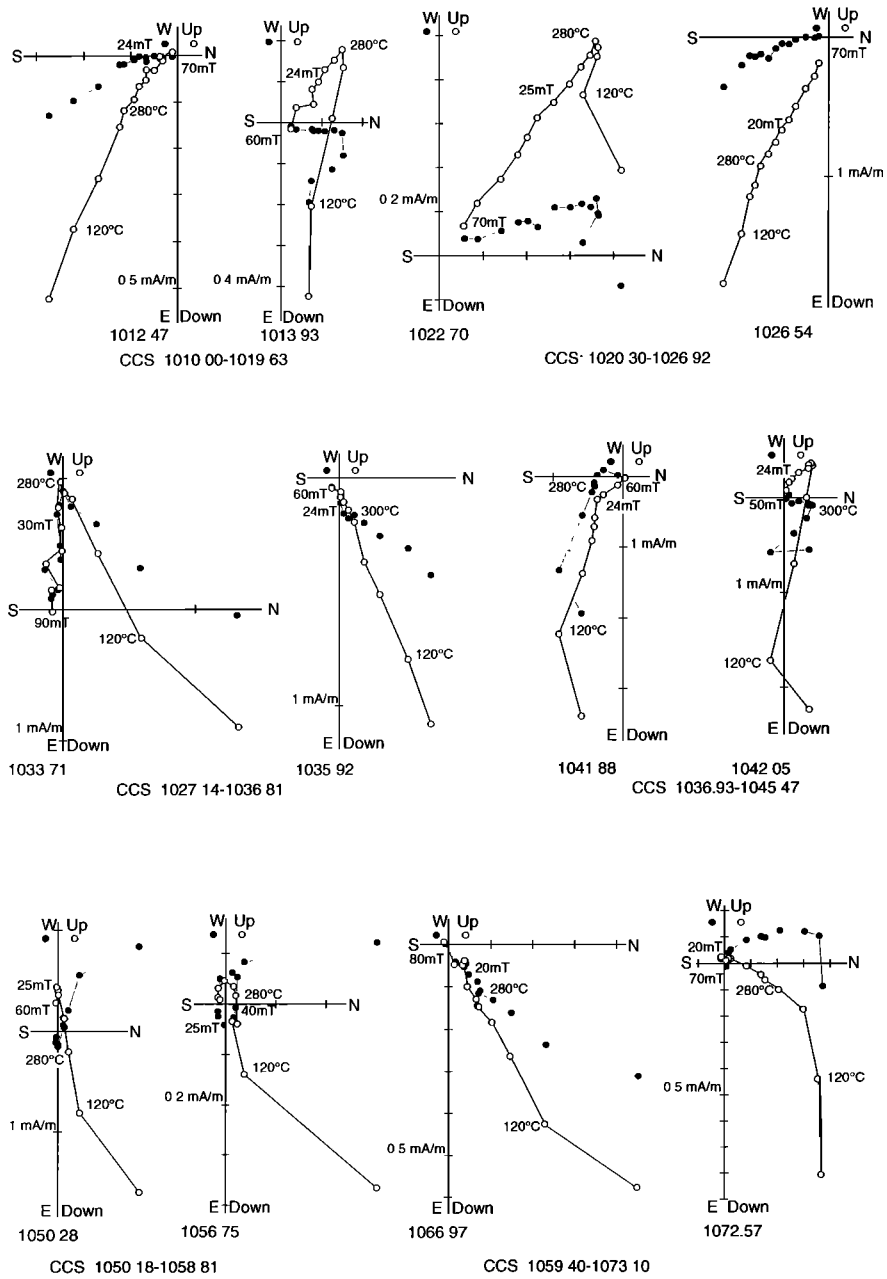


Figure 8. (continued)

good agreement with that of the expected present-day dipolar field at the drilling site (I = 67°) (Table 3 and Figure 9). This component is interpreted as a recent overprint acquired during the last 7×10^5 years (Brunhes Chron). Taking into account a thermal gradient of 3°C/100 m in the Paris Basin [*Vasseur*, 1982], any viscous remanence acquired during the last few million years under a maximal temperature of 50°C is surely removed during heating to 300°C [e.g., *Pullaiah et al.*, 1975].

The "higher-temperature" component isolated by AF technique between 5 and 90 mT (Figure 8) is characterized by both normal and reversed directions, which are antipodal within each CCS (Figure 10). This component is interpreted as the characteristic remanent magnetization (ChRM).

Because the inclinations of the present-day dipolar field at the drilling site and of the lower-temperature component are similar, the core can be restored with confidence in its initial (i.e., unrotated) orientation. Lower-temperature component mean direction is calculated for each CCS and is then rotated in order to coincide with the present-day dipole field direction (D = 0° and I ≈ 67°). For each CCS, the same correction is then applied to restore the ChRM mean direction.

As measured at 920 m depth, the drill string dips 84° toward 280° Under this coordinates system, with the drill string being nonvertical, the present-day dipole field direction is D = 14° and I = 67°. Therefore in this reference system the lower-temperature component must be rotated until it coincides with an azimuth of 14°. Once the data are put into the proper core reference frame, a correction for dip of the drill string (6° with a strike of 190°) can be made.

Figure 11 shows normal and reverse ChRM mean directions

Table 3. Depth and "Site-Mean" Lower-Temperature Component Directions From Each Composite Core Segment of the Montcornet Core

Depth, m	*N*	Dec	Inc	*K*	α_{95}
927 23-931.20	23	88.2	67 6	42.4	4.7
931.23-939.40	41	56 9	53 6	13.1	6.4
939.53-948 80	33	157.7	53 8	35.1	4.3
948.86-956 98	27	30.7	61.9	27.8	5.4
958.08-965.98	21	168 6	60 5	47.3	4.7
966 50-967.70	9	133 7	65.1	64.6	6.5
968.75-970.98	5	206.1	60 7	107 7	7.4
971.48-973.88	8	324 9	67 5	100.7	5.5
974 90-982.90	11	121.5	62.7	100.6	4.6
982.95-992.30	42	282.6	62 1	50.4	3.1
992.39-998 08	17	306 4	59.9	34.4	6 2
1000.69-1009.50	31	292 6	63 7	54.2	3 5
1010.00-1019.63	26	156.8	66.0	53.9	3 9
1020.30-1026.92	26	128.1	60.1	44 5	4.3
1027.14-1036 81	65	34.3	61.3	60.7	2.3
1036 93-1045 47	64	136.3	63 9	63.4	2.2
1045.86-1049.92	14	178 0	64.7	53.5	5.5
1050 18-1058.81	34	285.0	65 5	65 3	3.1
1059.40-1073.10	38	38 0	60 2	85.2	2.5
1073.16-1073.69	5	349.5	61.7	55.9	10.3
1074.17-1074.50	3	162.6	66.9	162 4	9.7

N, Number of samples, Dec, declination; Inc, inclination; *K*, precision parameter; and α_{95}, radius of the circle of 95% confidence.

calculated for each CCS (Table 4), after correction of the azimuth orientation of the drill core into core coordinates. It is clear that the two clusters of ChRM mean directions are antipodal (D = 44.9°, I = 51.6°, and α_{95} = 3.1° for normal directions; D = 230.7°, I = -50.5°, and α_{95} = 5.2° for reversed directions). Application of vertical core correction to the total mean (D = 45.9°, I = 50.4°, and α_{95} = 3.1°) determined from means of CCS intervals then yields D = 39.4°, I = 53.6°, and α_{95}=3.1° in geographic coordinates.

Even for samples with a clay content up to 80%, magnetic anisotropy measurements suggest that the error margin of ChRM inclination is not significant (less than 3°) and that the assumption of a rectilinar core is valid. It is then legitimate to calculate a virtual geographic paleopole (longitude is 107.4° and latitude is 57.3°). Magnetic anisotropy data and tectonic implications of this paleopole will be presented in a separate paper

Special Cases

Although characterized by a consistent inclination of about 65°, the lower-temperature component direction of a few samples does deviate from that of CCSs. It appears that the latter specimens come from very short core segments sandwiched between two CCSs. The different values in declination suggest that the rock of these small core segments underwent differential rotation with respect to the main core coordinates. As a consequence, these samples are not taken into account when calculating mean directions of core segments. However, they still do provide valuable information about magnetic polarity.

The ChRM of two groups of samples, three specimens each, could not be calculated by means of least square regression. Successive ChRM directions of these samples seem to fit a great circle with concomitant change from normal to reverse polarity (Figure 12). As these samples belong to the same CCS, the reversed ChRM direction could be assessed from the intersection of great circles [*McFadden and McElhinny*, 1988]. The corresponding inclinations are shown by ellipses in Figure 13.

Magnetostratigraphic Results

ChRM inclination, bulk susceptibility, resistivity, and gamma ray logs, as well as lithology are plotted in Figure 13 as a function of the core depth. Variations in values of bulk susceptibility, resistivity, and gamma ray logs evidently reflect the changing properties of the lithological succession. In contrast, absence of correlation between the fluctuations of these parameters and the changes in magnetic polarity provides an additional indication which further strengthens the primary origin of the ChRM.

Eighty-one polarity intervals are recognized in this study. The magnetic polarity zones are defined by ChRM directions from more than one sample, except for nine zones which are established by a single sample only. The two reverse polarity intervals based on single samples are not equivocal because these samples have an obvious present-day overprint removed at 300°C. The validity of the seven normal polarity intervals based on single samples is supported by the presence of well-defined ChRM after heating to 300°C (Figure 8, sample 988.25, CCS 982.95-992.30).

On the basis of the available biochronologic control, the presence of 35 polarity intervals appears as a minimum estimate for the Sinemurian part of the core (from 927 to 1023 m depth). Assuming that the early/late Sinemurian boundary occurs between 987 and 992 m depth as proposed by P. Andreieff (unpublished report, 1991) (Figure 3), the early Sinemurian contains polarities that are dominantly reversed

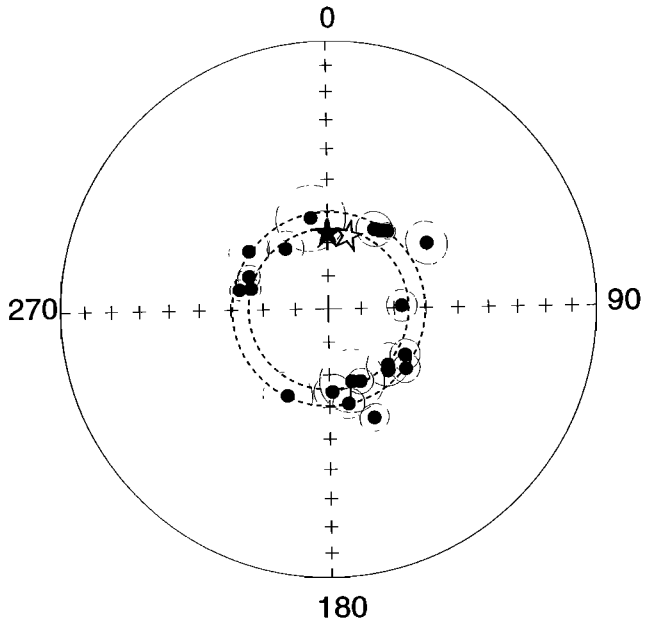


Figure 9. Equal-area projection of the lower-temperature component means per CCS. Solid and open stars represent the dipolar earth field projections before and after vertical correction, respectively, both in core coordinates. Small circles I = 60° and 65° are drawn for comparison.

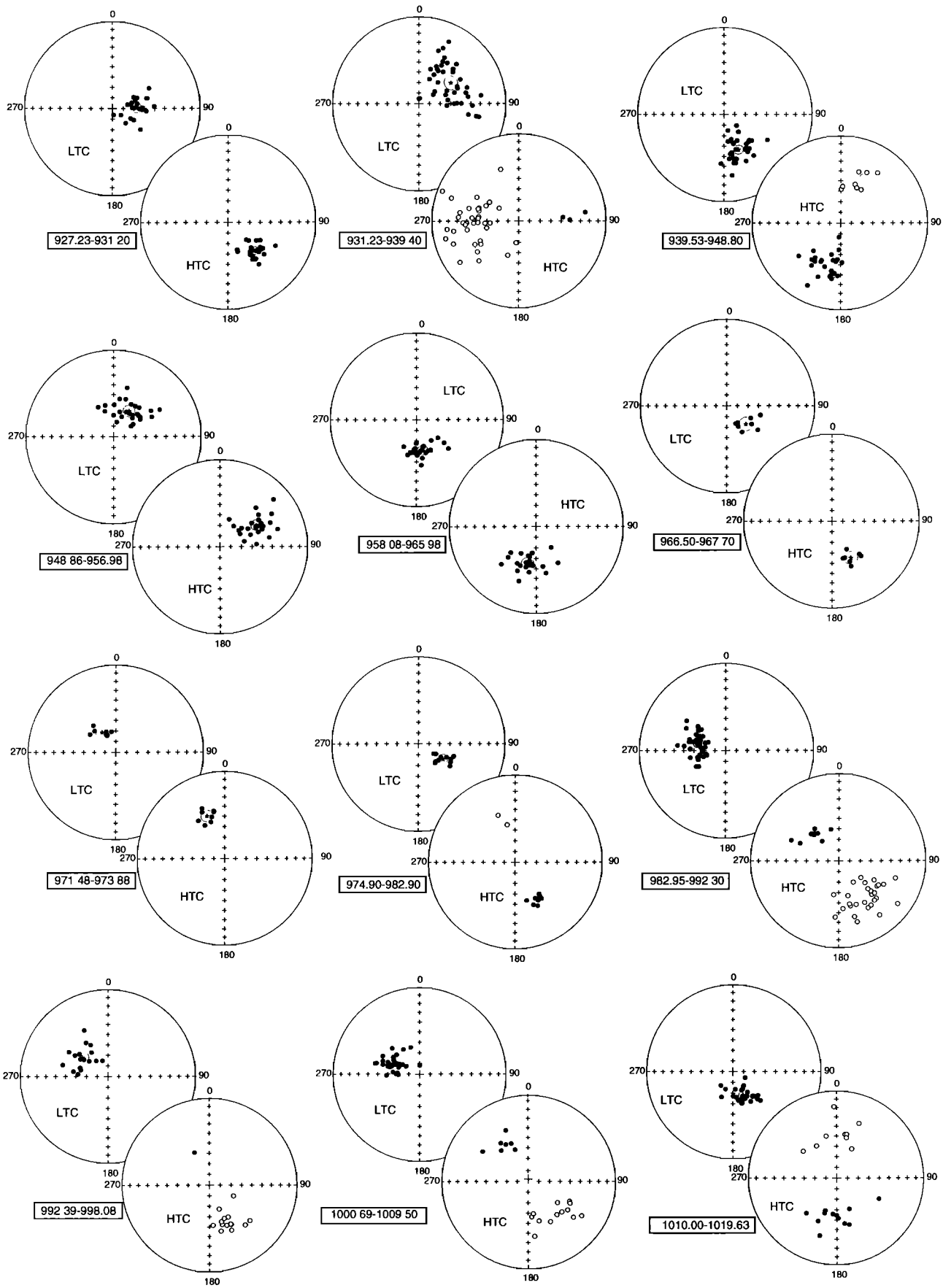


Figure 10. Equal-area projection of the lower-temperature component (LTC) and the higher-temperature components (HTC) for each composite core segment. Solid (open) symbols refer to the down (up) directions (Table 4).

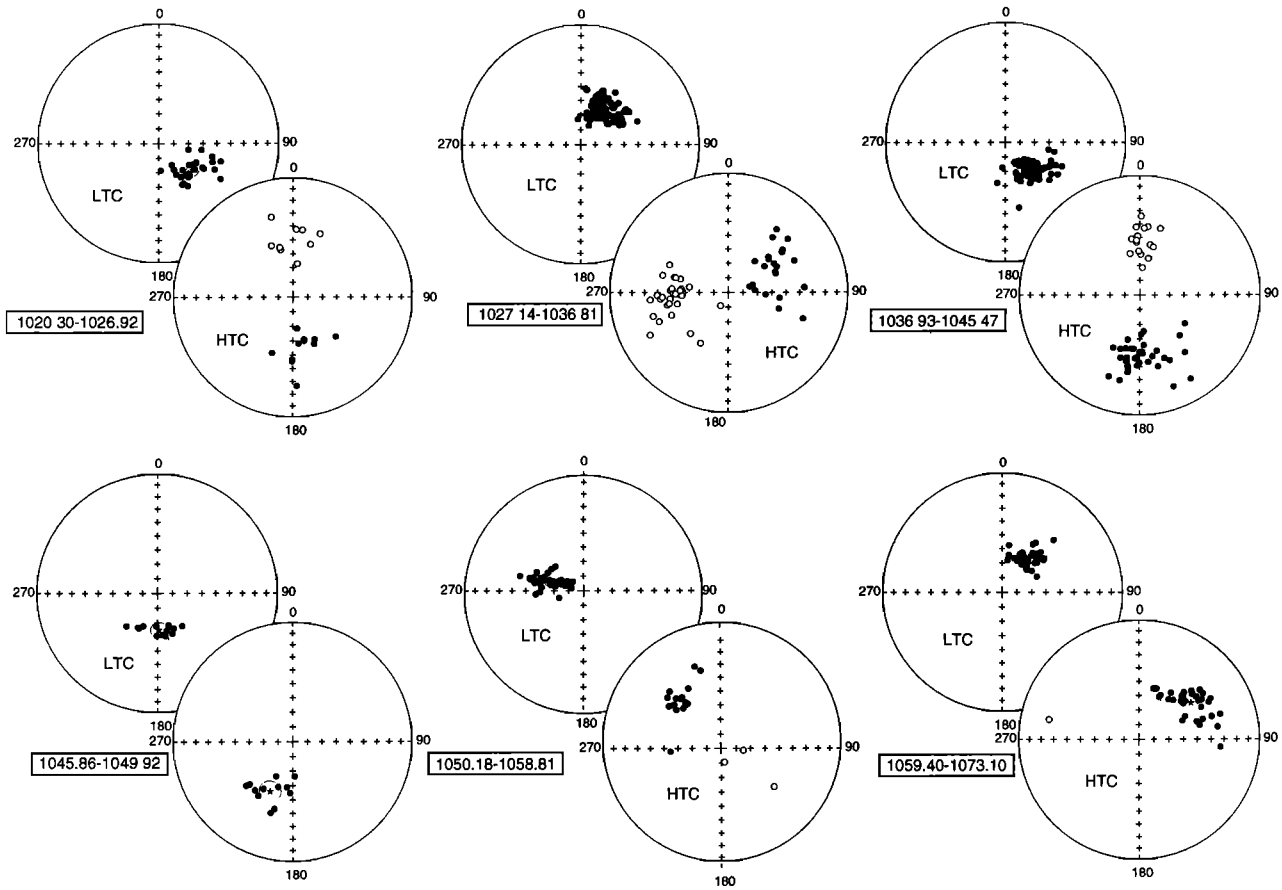


Figure 10. (continued)

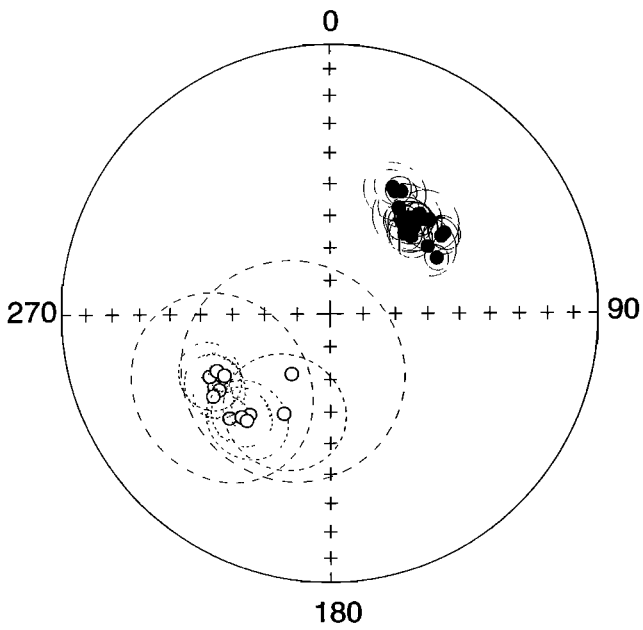


Figure 11. Equal-area projection of mean characteristic remanent magnetization (ChRM) directions corrected for mean lower-temperature component from each composite core segment (CCS), same convention as in Figure 7b.

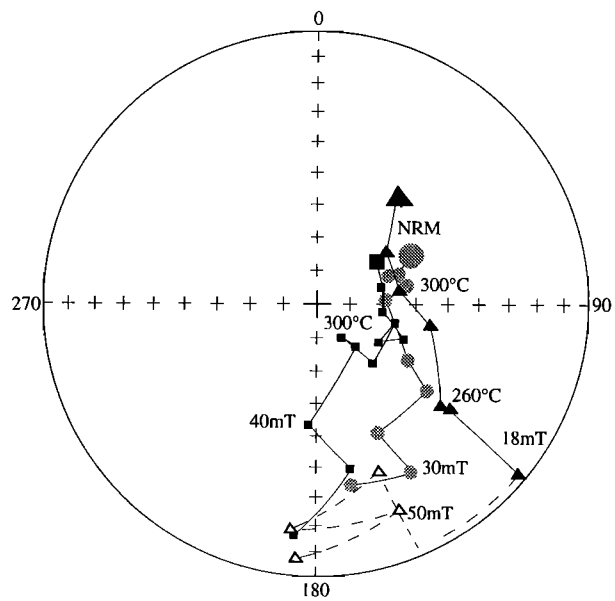


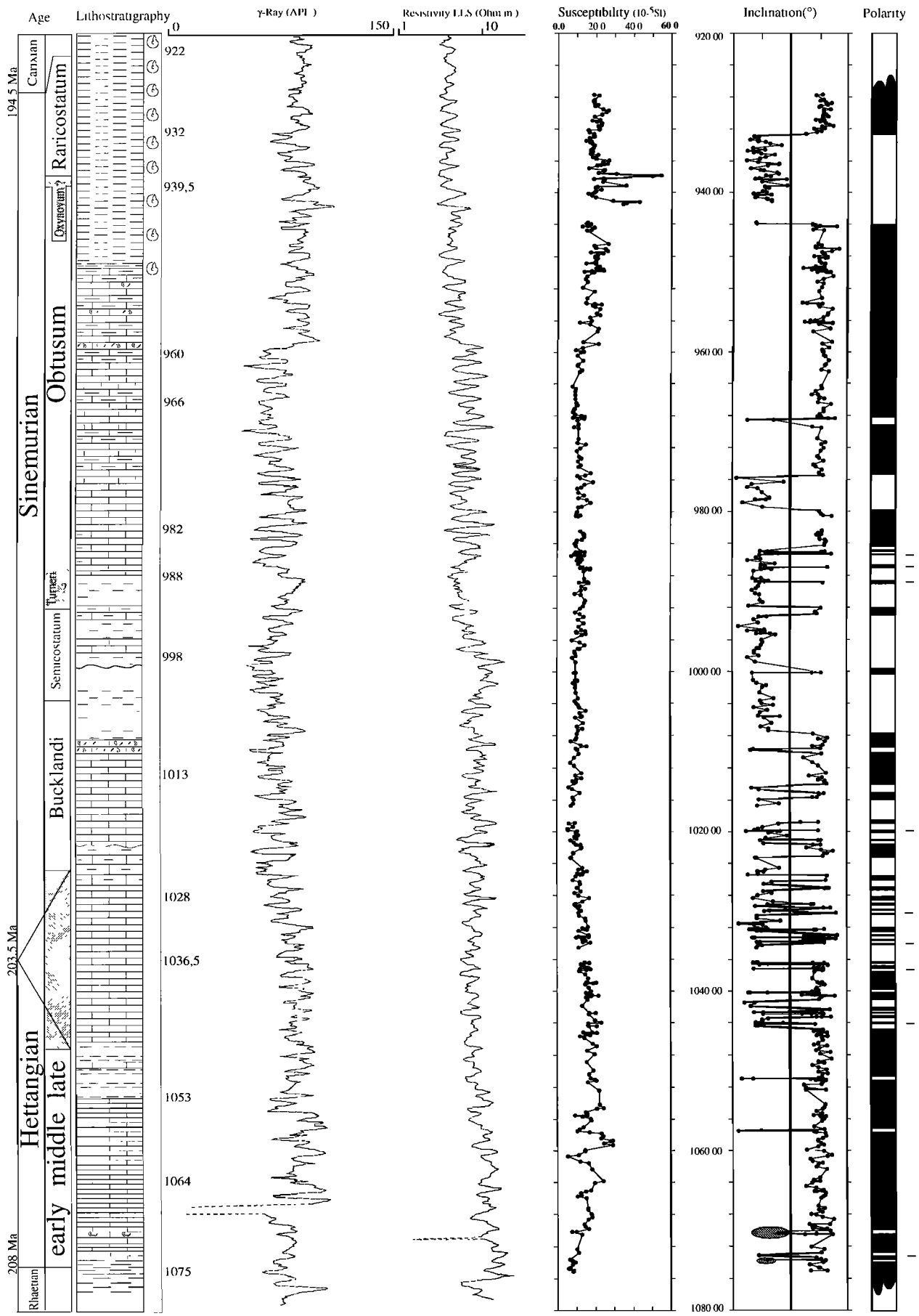
Figure 12. Equal-area projection of thermal (°C) and AF (mT) demagnetizations of three successive samples between 1070.24 and 1070.42 m is shown by circles shaded, squares and triangles, same convention as in Figure 7b.

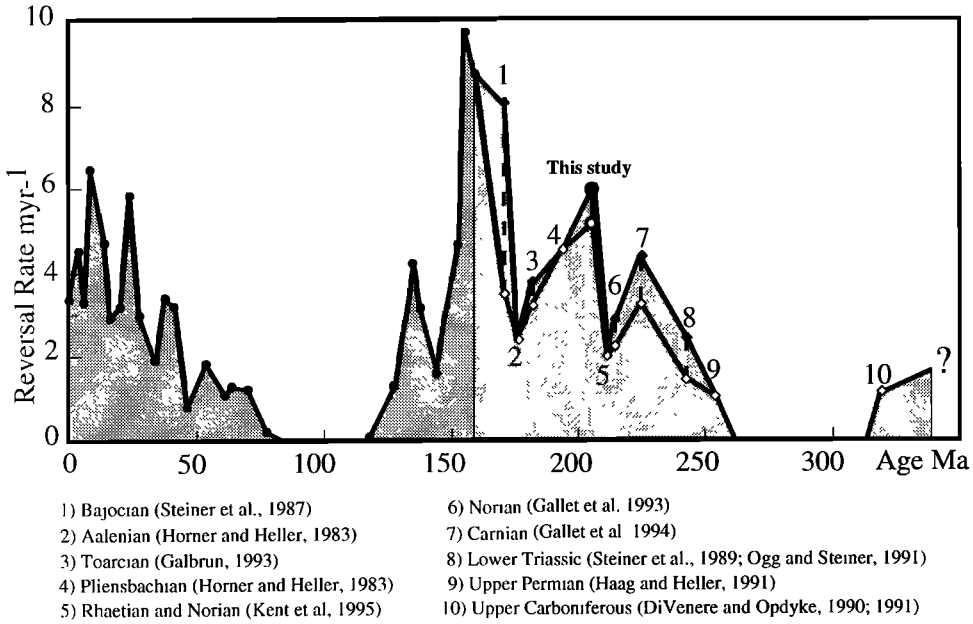
Table 4. Segment-Mean Declinations of the Lower-Temperature Component (LTC) and Directions of the Components From Alternating Field Demagnetization After Heating to 300°C (HTC) in Core Coordinates and Segment-Means Directions of HTC After Reorientation of the LTC Direction to Dipolar Field Direction (D = 14°, I = 67°) Before Vertical Correction of the Montcornet Core

Depth, m	LTC	HTC		HTC After Reorientation			
	Dec	Dec	*N*	Dec	Inc	*K*	α_{95}
927.23-931 20	88 2	136 9	23	62.7	52.9	67.9	3.7
931 23-939 40	56 9	267 2	41	224.3	-44.9	12.3	6.6
939 53-948.80	157.7	199 8	33	56.1	47.9	32.0	4.5
		199 1	25	55.4	48.0	29.9	5.4
		21 8	8	238.1	-47.6	36.0	9.4
948.86-956 98	30.7	59.8	27	43.1	50.4	30.6	5.1
958.08-965.98	168.6	193.2	22	38.6	54.2	32.1	5.6
966.50-982.90			31	33.1	48.5	195.0	6.6
966.50-967 70*	133.7	152.9	7	33.2	51.1	100.0	6.1
968.75-970 98*	206.1	220.4	5	28.3	47.2	72.4	9.1
971.48-973 88*	324.9	337.6	8	26.7	46.3	104.5	5.4
974.90-982 90*	121.5	152 2	11	44.7	48.7	147.4	3.6
		149 8	9	42 3	48.6	222.3	3 5
		343.2	2	235.7	-48.3		
982.95-992 30	282.6	321.8	39	53.2	48.2	22.1	5.0
		315 2	10	46.6	55.6	56.5	6.5
		143 7	29	235.1	-45.5	19.8	6.2
992 39-998 08	306.4	151 5	16	219.1	-50.5	43.2	5.7
		151.3	15	218.9	-50.1	41.1	6.0
		335 5	1	43.1	56.3		
1000 69-1009 50	292.6	322 1	20	43.5	47.8	27.4	6.3
		325.6	7	47 0	47.9	45.1	9.1
		140 1	13	221,5	-47.8	21.8	9.1
1010 00-1019.63	156.8	181.6	22	38.8	50.2	18.3	7.4
		182.0	12	39 2	52.1	18.5	10.4
		1.1	10	218.3	-48.0	16.7	12.2
1020.30-1026.92	128.1	173 2	20	59.1	51.0	27.9	6.3
		169 9	11	55.8	53.5	27.7	8.8
		356 8	9	242.7	-47.9	28.1	9.9
1027.14-1036.81	34 3	76 2	52	55.9	52.3	20.9	4.4
		64 0	22	43.7	52.9	19.0	7.5
		263.8	30	243.5	-50.8	27.0	5.2
1036.93-1045.47	136.3	178 8	53	56.5	48 4	28.3	3.7
		177 3	36	55.0	46.6	23.8	5.0
		2 3	17	240.0	-52.2	54.8	4.9
1045 86-1049 92	178.0	204.7	12	40 7	52.5	38.2	7 1
1050.18-1058.81	285 0	311 4	21	40.4	53 8	32.7	5.6
		312 0	18	41.0	51.4	47.2	5.1
		124	3	213.0	-68.2	14 0	34.2
1059 40-1073.10	38 0	54 6	33	30.6	46.3	30.0	4.6
1073 16-1073.7†	349 5	322 1	4	332.6	49.9	28.6	17.5
1074.17-1074.5†	162.6	182.9	3	34 3	49.8	85.5	13.4
Mean of Normal polarity directions			15	44.9	51.6	155.9	3.1
Mean of Reversal polarity directions			11	230.7	-50.5	76.6	5.2
Mean between 927 and 1020			10	45.2	49.9	156.6	3.9
Mean between 1020 and 1073			6	47.1	51.2	104.7	6.6
Total mean			16	45.9	50 4	130 1	3.1

Sites were grouped as one segment mean.
† Sites were excluded for total mean.

Figure 13. (opposite) Magnetostratigraphy and susceptibility of the Hettangian and Sinemurian from the Moncornet drill cord (Paris Basin). Depth scale is in meters. Relative age is obtained from ammonoid occurrences where indicated with corresponding symbol. Other relative ages are inferred from correlations with foraminiferas, ostracods, pollens, and sequence stratigraphy interpreted from γ ray and resistivity logs (from ANDRA, unpublished report, 1994; R. Jan du Chêne 1994, unpublished report, 1994). Unidentified ammonoid zones and uncertain ages are shaded. Absolute time scale after *Harland et al.* [1990]. Lithology, γ ray log (American Petroleum Institute), and resistivity log (Laterolog Long Spacing, ohm x m) after *Trouiller et al.* [1992]. Note that polarity intervals defined by single samples are identified with ticks.


Age
Lithostratigraphy
γ-Ray (API)
0
150
Resistivity LLS (Ohm m)
1
10
Susceptibility (10-5SI)
0.0
20.0
40.0
60.0
Inclination(°)
Polarity
194.5 Ma
203.5 Ma
208 Ma
Sinemurian
Hettangian
Carixian
Raricostatum
Oxynotum?
Obtusum
Turneri
Semicostatum
Bucklandi
late
middle
early
Rhaetian
922
932
939,5
960
966
982
988
998
1013
1028
1036,5
1053
1064
1075
920.00
940.00
960.00
980.00
1000.00
1020.00
1040.00
1060.00
1080.00



1) Bajocian (Steiner et al., 1987)
2) Aalenian (Horner and Heller, 1983)
3) Toarcian (Galbrun, 1993)
4) Pliensbachian (Horner and Heller, 1983)
5) Rhaetian and Norian (Kent et al, 1995)
6) Norian (Gallet et al. 1993)
7) Carnian (Gallet et al 1994)
8) Lower Triassic (Steiner et al., 1989; Ogg and Steiner, 1991)
9) Upper Permian (Haag and Heller, 1991)
10) Upper Carboniferous (DiVenere and Opdyke, 1990; 1991)

Figure 14. Changes in the magnetic reversal frequency since 350 m.y. (modified from *Gallet et al.* [1992]). The new estimate of mean reversal rate (five reversals per million years) is based on Hettangian-Sinemurian stage duration by *Harland et al.* [1990]. Changes in reversal frequency after 160 m.y. are obtained from the magnetic timescale of *Harland et al.* [1990], using a 10 m.y. mobile mean with 5 m.y. translation steps [after *Gallet et al.*, 1992]. Estimation of mean reversal rates of the late Permian-Bajocian time span with (without) polarity intervals defined by single samples indicated by solid (open) diamonds, respectively.

with a relatively high reversal frequency. Conversely, the late Sinemurian emerges as a normal-dominated polarity time span, with a comparatively low reversal frequency.

The strata from 1025 to 1053.5 m depth encompassing the Hettangian/Sinemurian boundary contains 40 polarity intervals. The underlying Hettangian (from 1053.5 to 1074.5 m depth) is essentially characterized by mainly normal polarity. A minimum of five reversed polarity intervals of short duration are recognized in the Hettangian.

Discussion and Comparison

The conclusions from the rock magnetic study suggest that magnetite is the dominant magnetic carrier. Absence of correlation between polarity reversals and changes in magnetic susceptibility, resistivity, and gamma rays further substantiate the primary origin of the ChRM. As we are dealing with a relatively expanded sedimentary sequence, coupled with an acceptable, but still slightly imprecise, biochronological control, the component presently recognized as a ChRM represents a reliable record of the magnetic polarity sequence. To date, the magnetostratigraphic sequence obtained from this epicontinental marine series in Paris Basin is the most complete ever documented for earliest Jurassic times and meets most criteria required for a magnetostratigraphic standard.

The Hettangian stage is clearly dominated by normal polarity. This is in agreement with previous results obtained from the nonmarine series interlayered with igneous extrusives of the Newark Basin [*Witte and Kent*, 1990; *Witte et al.*, 1991; *Kent et al.*, 1995]. *Olsen and Fedosh* [1988] and *Witte et al.* [1991] suggested a short time span (1 m.y.) for the entire Newark extrusive zone. According to *Fowell et al.* [1994], the Triassic/Jurassic boundary in the Newark Basin is well constrained by palynological and radiochronological data. Despite these chronological controls, comparison and correlation of the Hettangian magnetostratigraphy between Newark and Paris Basins remains problematic. Provincialism of palynofloras also impedes the construction of an independent correlation means for comparison of polarity sequences from the two basins. Another difficulty is that the Hettangian/Sinemurian boundary is not known from the nonmarine record of the Newark. However, both data sets support a predominantly normal polarity for the Hettangian stage thus underlining the contrast in reversal frequency with adjoining stages. This magnetostratigraphic signal may thus be used as a potential time marker for recognition of the Hettangian stage.

The normal polarity-dominated, low reversal frequency interval of the early Hettangian is followed by a high reversal frequency interval ranging from middle to late Hettangian to early Sinemurian. This interval of high reversal frequency does not seem to be recorded in the contemporaneous Umbrian Apennines series, which were found to have a total of six polarity intervals [*Channell et al.*, 1984]. Such a discrepancy may be explained by the combination of low sedimentation rates, gaps, and insufficient biochronological control of the Umbrian record.

Because of remaining uncertainties related to the exact position of stage boundaries in the Montcornet core, changes in magnetic reversal frequency during the Hettangian and Sinemurian stages can only be estimated. Using the *Harland et al.* [1990] absolute timescale, the 81 normal and reversed polarity intervals yield a relatively high mean of five reversals per m.y. for the two stages.

On the basis of the Global Paleomagnetic Database [*McElhinny and Lock*, 1991, 1993a, 1993b], *Johnson et al.* [1995] constructed a model of long term magnetic polarity reversal behavior for the last 570 m.y. and suggested an approximate period of 200 m.y. between superchrons during the Phanerozoic. *Gallet et al.* [1992] reviewed the available magnetostratigraphic data set covering the late Carboniferous to middle Jurassic time span and the changes of the magnetic reversal frequency since 350 m.y. Despite the error associated with the completeness and accuracy of the record, the estimated frequency of five reversals per million years for the Hettangian and Sinemurian stages improves the convergence in frequency pattern between the Permian-Bajocian (≈100 m.y.) and the Cretaceous-Recent time spans (≈83 m.y.). Therefore our results support a 150-200 m.y. time constant in the reversal process (Figure 14) for the last 350 m.y. [*McFadden and Merrill* ,1984; *Courtillot and Besse*, 1987; *Gallet et al.*, 1992].

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